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University Roll No . :

First Term Examination , Odd Semester 2015-16

B.Tech. (2nd Year)- III Semester

Subject:- Mathematics - III **Subject Code:- AHM-3001**

Time: 1 ½ Hours **Max. Marks: 20**

Section-A

Note: ***Attempt All Questions*** **:** **1×5=5**

1. Eliminate arbitrary constants a *and* b from the equation $\log(az-1) = x + ay + b$, to form a PDE.
2. Form a PDE by eliminating arbitrary function f and g from the equation $z = f(x^2 + y) + g(x^2 - y)$.
3. Solve $x^2p + y^2q = (x+y)z$.
4. Find the solution of $\left(D_x^3 - D_y^3\right)u = 0$.
5. Classify the PDE $z_{xx} + 2xz_{xy} + (1-y^2)z_{yy} = 0$, where $x^2 + y^2 = 1$.

Section –B

Note: ***Attempt Any Three Questions*** **:** **2×3=6**

1. Form a PDE by eliminating arbitrary function F from the equation $F(x^2 + y^2 - z^2, lx + my + nz) = 0$.
2. Solve $xy^2p - y^3q + axz = 0$.
3. Solve $r - 6s + 9t = \tan(y + 3x)$.
4. Solve by the method of separation of variable $\frac{\partial^2 u}{\partial x^2} - 2\frac{\partial u}{\partial x} + \frac{\partial u}{\partial y} = 0$.

PTO

Section – C

Note: ***Attempt Any Three Questions :*** **3×3=9**

1. Solve $(x^2 - y^2 - yz)\frac{\partial z}{\partial x} + (x^2 - y^2 - zx)\frac{\partial z}{\partial y} = z(x - y)$
2. Solve the PDE $(D - 3D' - 2)(D - D'^2)z = 6e^{2x}\sin(3x + y)$.
3. Find the complete solution of the one dimensional wave equation of the vibration of a string fixed at both end.
4. The ends A and B of a rod 20 cm. long have the temperatures at 30^0c and at 80^0 c respectively, until steady state prevails. The temperature of the ends are changed to 40^0c and 60^0c resp. Find the temperature distribution in the rod at time t.

Printed Pages-2 University Roll No.....

B. Tech. II Year III Semester, First Term Examination, 2015-16

Data Structures & Algorithms (CSE 3001)

Time: 1.5 Hrs. **Total Marks: 20**

Notes:

1. All parts of a question should be answered at one place.
2. Answer should be brief and to-the-point and be supplemented with neat sketches.
3. Any missing or wrong data may be assumed suitably giving proper justification.
4. Figures on the right-hand side margin indicate full marks.

Section A

Note: Attempt All Questions **(1x5=5)**

I. Write the condition of overflow and underflow in Circular Queue.

II. Explain the condition of single node in Circular linked list and doubly linked list.

III. What is an abstract data type?

IV. Write the polish notation of (A + B) / (C – D) + E.

V. What is a non-linear data structure? Give examples.

Section B

Note: Attempt any Three Questions **(2x3=6)**

I. Discuss the advantages of linked list over arrays. Also discuss the disadvantages of using linked list. Explain the applications of linked list.

II. Explain the static and dynamic representations of stack. Compare malloc( ) with calloc( ).

III. Explain O-notation. Prove or disprove the following

a) O-notation is symmetric b) O-notation is transitive.

IV. Evaluate the given postfix expression

3 5 + 6 4 - * 4 1 - 2 ↑ +

Also write the equivalent infix and prefix expression.

Section C

Note: Attempt any Three Questions **(3x3=9)**

I. Write an algorithm to delete a particular key in a circular linked list.

II. Given an infix expression convert into an equivalent postfix expression.

(A + B) ^ C – (D * E) /F

Also explain the steps to convert the infix expression into a prefix expression.

III. Discuss the applications of stack and queues. Let us consider a circular queue of length 4. Following operations are requested. Show the different states of the queue under the processing of these operations.

1. Enqueue A	2.Enqueue B	3.Enqueue C
4. Enqueue D	5.Dequeue	6.Enqueue E
7. Dequeue	8.Enqueue F	9. Dequeue
10. Dequeue	11. Dequeue	12. Dequeue

IV. Discuss the linked list representation of priority queues. Also explain the algorithm to insert an element in priority queues using doubly linked list.

B.Tech. II Year, III Semester, Ist Term Examination, 2015-16
CSE-3002: Computer Organization

Time: $1\frac{1}{2}$ Hour **Total Marks: 20**

SECTION-A

Attempt ALL Questions. **[5x1 = 5 marks]**

I. Estimate how many Full adders, half adders and AND gates will be required to multiply 16 X16 array multiplier?
II. An 8 bit register contains the binary value 10011100. What is the register value after an arithmetic shift right and logical shift right operations?
III. Perform the arithmetic operation $[(-62)_{10}+(+50)_{10}]$ using 2's complement method.
IV. Encode 01001110110 using the Hamming code and even parity.
V. What is the Gray equivalent of $(75)_{10}$.

SECTION –B

Attempt any THREE questions. **[3x2 = 6 marks]**

I. Explain IEEE Format for single and double precision floating point. Convert the decimal number -307.1875 in single and double precision format.
II. Design an adder/subtractor circuit with one selection variable S and two inputs A and B. When S=0, the circuit performs A+B. When S=1 the circuit perform A-B by taking 2's complement of B.
III. Implement half adder using NAND and NOR gates only.
IV. Find the decimal equivalent of following 32 bit floating point numbers
(a) A3358000(in hex) **(b)** BE580000 (in hex)

SECTION –C

Attempt any THREE questions. **[3 x 3 = 9 marks]**

I. Using only four-bit adders, construct an eight-bit adder. Each four-bit adder has two four-bit inputs and one five-bit output. Your eight-bit adder should have two eight-bit inputs and a one eight-bit output (ignore 9th output bit).
II. Draw and explain the organization of typical CPU with general purpose register organization.
III. Represent Booth's algorithm to multiply two binary numbers in flow chart form. Apply the Booth's algorithm to multiply (-9*6).
IV. Show the design of a 4-bit Carry Look-Ahead Adder (CLA) by drawing its logic diagram.

First Term Examination, 2015-16
Course: B.Tech. (CSE), Year:II, Semester:III
Database Management System (CSE 3003)

Uni. Roll No: _____

Time: 1.5 Hrs. **Total Marks: 20**

Notes:

1. All parts of a question should be answered at one place.
2. Answer should be brief and to-the-point and be supplemented with neat sketches.
3. Any missing or wrong data may be assumed suitably giving proper justification.
4. Figures on the right-hand side margin indicate full marks.

Section A

Note: Attempt All Questions **(1x5=5)**

I. Draw three schema architecture of database management system.

II. What are the constraints involved in the relational model of DBMS?

III. Find out union and intersection of the tables given below:

R1

X	Y	Z
a	1	4
b	2	5
b	2	9

R2

X	Y	Z
a	2	4
b	2	9
c	5	8

IV. What is left outer join? Give an example.

V. What do you understand by the term mapping cardinality? What are types of mapping cardinality?

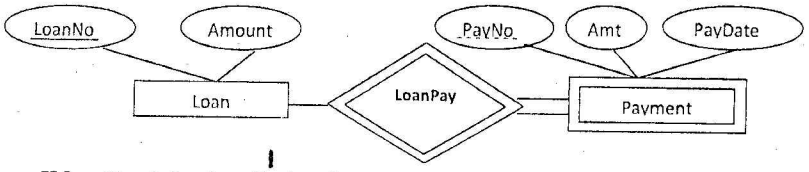
Section B

Note: Attempt any Three Questions **(2x3=6)**

I. What are four main differences between a file-processing system and a DBMS?

II. Explain the difference between physical and logical data independence.

III. Convert the following ER diagram to relations:



IV. Explain the distinction among the terms primary key, candidate key and super key with example.

Section C

Note: Attempt any Three Questions **(3x3=9)**

I. Consider the following schema:

Employee (Eno, Ename, dob, designation, address, salary, age, dno, Dep_name)

Department (Dno, Dname, Dlocation)

Dependent (Dep_name, dob, Relationship)

Write the following queries in relational calculus:

a) Find all employees whose salary is more than 50,000.

b) Retrieve the birthdate and name of the employees whose age is less than 30.

c) Retrieve the name and address of all employees working for research department.

d) Find the names of employees who have no dependents.

II. Discuss the various types of database users.

III. A company database needs to store information about employees (identified by ssn, with salary and phone as attributes), departments (identified by dno, with dname and budget as attributes), and children of employees (with name and age as attributes). Employees work in departments; each department is managed by an employee; a child must be identified uniquely by name when the parent (who is an employee; assume that only one parent works for the company) is known. We are not interested in information about a child once the parent leaves the company.

Draw an ER diagram that captures this information.

IV. Write short notes on:

(i) Specialization and Generalization

(ii) Aggregation

Printed Pages: 7 **University Roll Number:**

First Term Exam-2015 **[A]**

Ethics & Values: AHE-3001

Time: 1 ½ Hours **Max. Marks: 20**

Note: There are four passages in this paper. The passages are not unseen and are based on course discussion. They help in your analytical problem solving. Carefully read the passages and answer the following questions having the multiple answers. Carefully fill all the information in OMR sheet: Neatly in the blocks. No partial marking.
Full marks (assigned) will be given for a complete answer that means all correct choices.

Section A

Passage 1 (2X1.5=3) (Attempt All Questions)

We understand society as a group of people who interact through shared rules & norms, traditions and rituals. They develop a kind of culture & civilization over a period of time. Society develops codes of conduct which in turn develop a common understanding to what we believe is right or wrong, good or bad, and what should be done and what not. This all gradual development gives birth to morality. So, moral systems always require a society, and without systems of shared rules & ritual we cannot have real & active societies.

Questions:

1) What is the practical worth of codes of conduct?
a) Codes of conduct develop common understanding in society.
b) The common understanding creates confusions in society.
c) Codes of conduct explain what is right and wrong.
d) Codes of conduct are necessary for morality.

2) Some persons do not care about morals and shared understanding. What can be the consequences of this kind of attitude?
a) This attitude would make such people immoral.
b) Such people cannot be good team leaders.
c) Such people will be right in being immoral.
d) Such attitude is not good for their well being & success.

Passage: 2 (4X1.5=6)

There was a thirsty crow. He was searching for water. He found a pitcher with a little water in it. The neck of the pitcher was too long and the water level too low. The crow could not reach the water with her beak. She saw many pebbles around. Suddenly, the crow had an idea. She picked the pebbles one by one and dropped them in the pitcher. And, the level of water slowly rose to the top. The crow drank it thirstily. She was very happy. Let's re-understand the story with new insights in the contemporary challenging situations. The application of smart management and professional attitude seem unavoidable in the current situations. Rationality is necessary, and it is to understand each and everything whatever we do in a causal manner: what is a cause and how does it produce certain effects. This is what we call organizing or developing professional attitude. Smartness and quickness is the key of success but smartness does not mean being diplomatic or cheating with someone. It is simply a play of rationality and critical thinking that reduces our time and energy on the one hand and increases the success rate on the other. There was a little water for the crow like we have got rare good opportunities in current time and for that there is tight competition, but the crow smartly made all the efforts by using his rationality to drink the water and some other thirsty birds would have just left the pitcher because of a little water. Think! Where you are and what are you following; the typical mob mentality or you have some smart guts to do the things differently and quickly? Success demands all around 360 degree active thinking. Problem solving takes birth once you acquire the 360 degree professional attitude. Try to find out the problems and use your critical mind to arrive at possible solutions. We need to be a great problem solver than just a problem finder.

3) What is the moral of the story: Thirsty Crow in current time?

a) We should always try to find pebbles.

b) We need to be smart and quick in our thinking.

c) Rationality & critical thinking increase our success.

d) We should always think about self interest.

4) What do mean by problem solving?

a) Only finding out a problem.

b) Working on a problem with critical thinking.

c) Finding the solutions with better reasons.

d) Just reporting the problem.

5) What does author mean by the phrase a little water?

a) The scarcity of good jobs.

b) The excess of opportunities.

c) The strong competition.

d) The abundance of jobs.

6) What do you understand by the 360 degree attitude?

a) It is thinking critically on all sides.

b) It is a problem solving attitude with active reasoning.

c) It is necessary for a good professional.

d) It is rigid professional behavior.

Section B

Passage: 3 (4X1.5=6)

Ethical Decision Making is composed of the ability to engage in ethical discourse with rationality, critical thinking, problem solving and decision making to analyze the life issues (problems of one's personal life, social life, and professional life). Moreover, there are three different approaches to handle the problems in a wider sense:1) Consequentialist approach, 2) Non-consequentialist approach , and 3) Virtue based approach. Consequentialist approach put emphasis on the results of the choices, actions and decisions (CAD: hereafter). Jeremy Bentham founded the theory namely, Utilitarianism. According to this theory we should decide about the ethics of any CAD on the grounds of utility and utility by the goodness, happiness and pleasures (GHP). The GHP factor becomes guideline for the positive cases, while in negative cases we should minimize the pain and unhappiness. This theory claims to be universal account of morality. It is naturalistic as it is derived from the human nature that people desire happiness and pleasure and they avoid sadness and

pains. On the other hand, the non-consequentialist approach maintains that the ethics of any CAD must be governed by the very nature of rules, principles and morals. This system was originally developed by Immanuel Kant and he calls it the Deontological system. This approach takes into account the two important aspects:1) Rationality and 2) the Sense of duty. We must act on our duties. And, the duties must be mutually satisfied by the two human capacities; they are rationality and autonomy (freedom). This theory claims to be universal account of morality like the utilitarianism. But, it's not naturalistic as it is based on human rationality and not grounded in the human experiences and emotions.

Questions:

7) Where do we use the problem solving and decision making?
a) In professional life only.
b) In both professional and social life.
c) It can be used in personal life as well.
d) Only in professional and social life.

8) What do you mean by the deontological systems?
a) Systems based on the sense of duty.
b) Systems based on the results.
c) Systems based on the nature of action.
d) Systems that follows rationality.

9) What is not right about the GHP factor?
a) It is increasing pain and reducing the pleasure.
b) It is about the unhappiness, pain & badness.
c) It is about reducing the pleasure and increasing the pain.
d) It is about following one's duties.

10) What is wrong about the CAD?
a) It is choice only.
b) It is choice, action & decision.
c) It is about decision only.
d) It is about action only.

Passage: 4 (5X1=5)

The Run Way Trolley Case is constructed hypothetically to understand the ethical decision making. Hypothetical cases involve some constructed situations in order to test one's real decision making and problem solving skills. In the first version of the trolley car case suppose you are the driver of a trolley car hurtling down the track at sixty miles an hour. Up ahead you see five workers standing on the track, tools in hand. You try to stop, but you cannot. The brakes do not work. You feel desperate, because you know that if you crash into these five workers, they will all die. Suddenly you notice a side track off to the right. There is a worker on the track, too, but only one. You realize that you can turn the trolley car onto the side track, killing the one worker, but saving the five. How would you respond in this situation as a driver? Majority would respond that they turn the trolley with the reason that it would spare more lives. Taken for granted that the worker on the side track is innocent majority would say five men on the main track are also innocent. And, the decision would be based on the reason that we should go for the better consequences. However, the obvious reaction seems questionable analyzing the decision on non-consequentialist approach. Well, decision will be the same but not grounded in the results. The decision is followed by the rationality and sense of duty as the driver's rationality and duty both allows him to turn the run way trolley to save five workers. The virtue approach will allow the same decision on the grounds of the virtues to save human kind. And, the doubt that how can a virtue oriented person kills a worker on the other track becomes meaningless as in the both non-consequentialist and virtue approach the decision maker has intention to save five workers and not to kill one. Moreover, if one worker survives then the decision maker would be the one of the happiest person. Moving to second version when trolley car is automatic and there is no side track. You are standing on the bridge as onlooker. Down the track comes a trolley, and at the end of the track are five workers. Once again breaks do not work and trolley car is about to crash into those five workers. You feel helpless to prevent this disaster. You think jump on the track but finding yourself too thin to stop the trolley you change the mind. However, you notice alongside you a very fat man just leaning onto the bridge. You reasonably think that if you give him a gentle push he will fall on the track before the trolley car and will obstruct the track anyhow. Although the fat man will die, he would save the life of five workers. Majority would decline the choice to give a gentle push to the fat man for saving five down on the track. The choice is

not like the first version because this version is a bit open and free in that sense the onlooker has got option not to involve himself. In this version the onlooker is not directly in the situation like the driver in the first case. But, why the majority is not minding the better consequences in this version because if we compare the consequences they are same five lives saved at the cost of one. Well, we can say, in the second version the decision is not save five workers some how makes an impression that we commit moral killing of five worker by not pushing the fat man. Is letting five workers not equal to killing? We can argue that it is not as we are not the source of their death. But, we can say majority would not push but will have some guilt that they let five workers die. Analyzing from different approaches, we can say a non consequentialist would not push as it is not his duty. A virtue oriented person would do the same as it is wrong to push an innocent and any virtuous person cannot have such a character to kill someone. Although it seems that a consequentialist person would push the fat man but in fact he would not as innocent killing would threaten the society and make all the people insecure. And, it all would diminish the GHP factor of society. So, better he will be in his silence to this case. However, the third version of this case in which there is a change in terms of condition that we need not to push the fat man instead we are given a button that is operating a trap onto which the fat man is standing. We have to press the button and the trap will open and fat man will fall down on the track before the trolley car and would save the five men. Majority cannot be so reluctant and passive in this case as it was in pushing the fat man. As a result, majority will be divided in pressing or not pressing the button. What is the reason that actually divided the majority and allowed more number of persons to save five men by pressing the button? The reason for the person, who goes with courage and applies a wider reason beyond his own interest of saving more people, is that the third version involves less guilty factor than the second, because if a person has this compulsive intention to kill one man in all the three versions then the person has more reasonable and wider intention to save the five. The factor of guilt is decisive for the person who really wants to save five. And, the person who really saves the five in all three versions would have less guilty sense in pressing the button comparing to push the fat man. Killing fat man or letting five workers die would make the equal guilt sense. Moral killing makes sense when we analyze the three versions on guilt factor. This hypothetical case challenges our ordinary pattern of reasoning and somewhere proved that our ordinary reasons proved to be wrong when assessed through critical thinking.

Questions:

11) Why majority would is in favor to turn the trolley in the first version?
   a) Saving five lives is better decision.
   b) We should take the decisions which can increase GHP factor.
   c) Majority find it right to kill one innocent on the other track.
   d) Majority has more sympathy to 5 workers.

12) What do you mean be hypothetical cases?
   a) They are based on real condition.
   b) They are based on constructed conditions.
   c) They are helpful in understanding the real decisions.
   d) Runway trolley case is an example of hypothetical case.

13) What are the practical insights of this case?
   a) Our ordinary rationality can be wrong.
   b) Critical thinking reveals better reasons.
   c) Innocent killing cannot be defended.
   d) It challenges our thinking and rationality.

14) Which of the following statements are true about this case?
   a) It is a hypothetical case.
   b) Majority would turn the trolley in second version.
   c) Majority would not push the fat man in second version.
   d) Majority would turn the trolley in the first version.

15) What is guilty sense?
   a) It is realization of wrongness.
   b) It is self understanding of one's wrong acts.
   c) It is forced realization.
   d) It is a good feeling.

B.Tech. II Year, III Semester, Ist Mid Term Examination

MBA-5031: Industrial Economics

Time: 90 Mins. **Total Marks: 20**

Section-A

Attempt All Questions **[5×1 = 5 Marks]**

1. State two features of resources that give rise to an economic problem.
2. What do you mean by "Demand"?
3. What happens to total expenditure on a commodity when its price falls and Price Elasticity of Demand is more than unit elastic?
4. What will be the slope of demand curve in case of Negative Income Elasticity of Demand?
5. Briefly discuss Deductive Methodology.

Section- B

Attempt any THREE Questions **[3×2 = 6 Marks]**

1. Define Economics. Explain its importance.
2. What do you mean by Elasticity of Demand? What are its types? Briefly discuss.
3. Differentiate between Static Economics and Dynamic Economics.
4. Discuss relation amongst Science, Engineering and Economics.

Section- C

Attempt any THREE Questions **[3×3 = 9 Marks]**

1. Briefly discuss the nature and scope of economics.
2. Define Law of Demand with Its assumptions and exceptions.
3. Explain the "law of diminishing marginal utility" with a suitable example. Discuss its exceptions.
4. Mr. X spends as follows for a commodity at various prices:

Price per Kg. (Rs.)	Total Outlay (Rs.)
8	9000
7	10500
6	12000
5	14000

Find out the elasticity of demand for the case: **(i)** When the price decreases from Rs. 8 to Rs. 7, & **(ii)** When the price increases from Rs. 6 to Rs. 8

Set-A

University Roll No._____

GLA UNIVERSITY, MATHURA

B. Tech. II Year (Odd Semester), I Term Examination, 2015-16

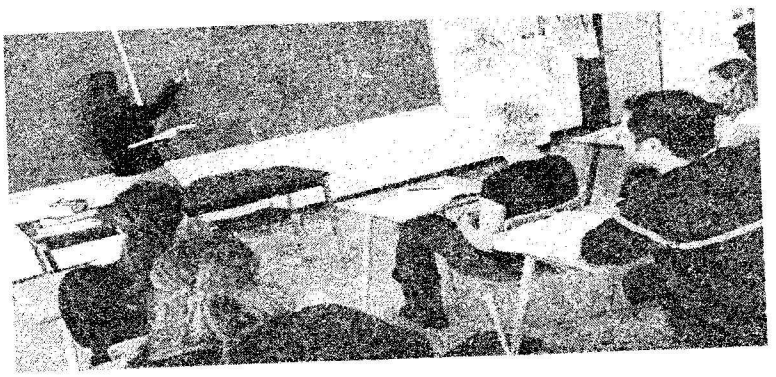
Subject: Business English
Time: 90 Minutes
Max. Marks: 15

Note:

- Answer all questions from Section A, any 3 out of 4 from Section B, and any 2 out of 3 from Section C.
- Answers should be to-the-point, and word limit for informative questions from Section B (approx 60 words) and Section C (150 words) should be maintained.
- All questions of a particular section should be answered at one place.
- Figures on the right-hand margin indicate marks.

Section A **0.5 x 6=3**

1. Put stress on the appropriate syllables in the underlined word: My friend gave me a wonderful present.
2. Identify the intonation in the following utterance:
   Have you completed your homework?
3. What does the gesture of "hands on cheek during a lecture" generally mean?
4. You go to meet the chief warden regarding some problems in your hostel. Name the flow of communication involved in this process.
5. The vice-chancellor of your university addresses the audience on the Independence Day. Which level of communication will that be?
6. Look at the picture and identify the proxemic zone:



Section B

2x3=6

1. You have a French friend visiting you at your place and you introduce her to your father. The girl wants to shake hand with your father which makes him feel uncomfortable. What kind of barrier to communication is at work here? How should it be removed?
2. What are the components of voice modulation? Write a short note either on Pause or Tone.
3. You ask your friend for a favour and receive a delayed response. Analyse the delay in terms of chronemic communication.
4. Identify the intonation pattern and purpose in the following sentences:
   a. Are you coming tomorrow?
      Intonation: Rising/ Falling/ Fall-Rise
      Purpose: Confirmation/ Asking for Confirmation
   b. I do not support US Military operation in Syria.
      Intonation: Rising/ Falling/ Fall-Rise
      Purpose: Statiag a fact/ stating a confusion

Section C

3x2=6

1. The Literary Club of your university is organising a debate competition. As the coordinator of the club, you have to invite the university authorities, faculty members, your classmates and your juniors. Discuss the process of communication with reference to the choice of medium, flow and level(s).
2. Write a note on the various proxemic zones with suitable examples.
3. Identify the syllables in the following words and mark the stressed syllable in each of them:

a. Botany	b. Botanical	c. University
d. Engineer	e. Photography	f. Examination

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